The global contract and the delegated contracts

27 October 2026

Ports

The planner M_1 reads $u = (s, h, g)$, the fence half-side $s \in \mathbb{R}$, the home $h \in \mathbb{R}^2$ and the goal $g \in \mathbb{R}^2$, and writes the mission $w = (w_1, w_2)$, two waypoints in $\mathbb{R}^2$. The controller M_2 reads the mission w and the environment $e = (p_0, dt)$, the start position $p_0 \in \mathbb{R}^2$ and the tick lengths $dt_0, \ldots, dt_8$, and writes one target $y_t \in \mathbb{R}^2$ at every tick $t = 0, \ldots, 8$. The planner never sees e or y; the controller never sees u. $\|\cdot\|_\infty$ is the maximum of the absolute coordinates and $\|\cdot\|_2$ the Euclidean norm.

Global contract (A, G)

The assumption A is a condition on (u, e) and the guarantee G a condition on (u, e, y).

$$A: \quad 25 \le s \le 50, \qquad \|h\|_\infty \le s - 3, \qquad \|g\|_\infty \le s + \tfrac{1}{2}, \qquad \|p_0 - h\|_\infty \le 1, \qquad dt_t \in [0.9,\, 1.1] \text{ for every } t.$$

$$G: \quad y_t \in [-s, s]^2 \text{ for every } t \qquad \text{and} \qquad \|y_t - g\|_2 \le 3 \text{ for some } t.$$

In words, the fence is at least 50 m across, home is 3 m inside it, the goal may be up to half a metre outside it, the vehicle starts within 1 m of home and ticks are close to one second; then every target stays inside the fence and some target comes within 3 m of the goal.

Delegated contracts (proposer A, verified)

Planner contract (A_1, G_1). A_1 is a condition on u and G_1 a condition on (u, w).

$$A_1: \quad 25 \le s \le 50, \qquad \|h\|_\infty \le s - 3, \qquad \|g\|_\infty \le s + \tfrac{1}{2}.$$

$$G_1: \quad 5 \le \|w_2 - w_1\|_\infty \le 100, \qquad \|w_1\|_\infty \le 100, \qquad \|w_2\|_\infty \le 100,$$
$$6 \le \|w_1 - h\|_\infty \le 99, \qquad \|h\|_\infty \le s - 1, \qquad \|w_1\|_\infty \le s, \qquad \|w_2\|_\infty \le s - 1, \qquad \|w_2 - g\|_2 \le 3.$$

The first line of G_1 makes the mission acceptable to the controller. The second line keeps the first leg valid from any start the global assumption allows, keeps the box the controller may sweep inside the fence, and places the last waypoint close enough to the goal.

Controller contract (A_2, G_2). A_2 is a condition on w and G_2 a condition on (w, e, y).

$$A_2: \quad 5 \le \|w_2 - w_1\|_\infty \le 100, \qquad \|w_1\|_\infty \le 100, \qquad \|w_2\|_\infty \le 100.$$

$$G_2: \quad \text{if} \quad 5 \le \|w_1 - p_0\|_\infty \le 100, \quad \|p_0\|_\infty \le 100, \quad dt_t \in [\tfrac{1}{2}, \tfrac{3}{2}] \text{ for every } t, \quad \text{then}$$
$$\min(p_{0,k}, w_{1,k}, w_{2,k} - 1) \le y_{t,k} \le \max(p_{0,k}, w_{1,k}, w_{2,k} + 1) \text{ for every } t \text{ and axis } k \in \{x, y\},$$
$$\text{and} \quad y_t = w_2 \text{ for some } t.$$

A_2 is exactly the controller's input guards. G_2 promises, for environments inside the code's clamps, that the target never leaves the box spanned by the start and the two waypoints, widened by 1 m around w_2 only, and that it reaches w_2 exactly. The widening comes from the handover code, which moves the target by less than half the waypoint radius at the blend tick.

Why these compose to (A, G)

D1	$A \subseteq A_1$	A_1 is the part of A about u.
D2	$G_1 \subseteq A_2$	the first line of G_1 is A_2.
D3	$A \cap G_1 \cap A_2 \cap G_2 \subseteq G$	the box of G_2 lies in the fence by the margins of G_1; $y_t = w_2$ gives $\|y_t - g\|_2 \le 3$.
R	$\forall u \in A_1 \; \exists w \; G_1(u, w)$	witness: planner R4 with margin 1.

All four were decided by Z3 (D3 in 5.7 s, the rest under 2 s). Then $M_1 \models (A_1, G_1)$ was checked on the planner code (0.6 s) and $M_2 \models (A_2, G_2)$ on the controller code by bounded model checking over 8 ticks with w, p_0 and dt free (38.7 s). By the composition rule, $M_1 \cap M_2 \models (A, G)$.

The other two verified pairs

Proposers B and C found the same A_1 and A_2 and the same environment condition in G_2. They differ in the box and the margins.

	box in G_2, and arrival	margins in G_1
A	box of p_0, w_1, w_2, widened by 1 around w_2; $y_t = w_2$	$\|h\|_\infty \le s-1$, $\|w_1\|_\infty \le s$, $\|w_2\|_\infty \le s-1$, $\|w_2-g\|_2 \le 3$
B	box of p_0, w_1, w_2 widened by 1 on all sides; $\|y_t - w_2\|_2 \le \frac{1}{2}$	$\|h\|_\infty \le s - \frac{9}{4}$, $\|w_1\|_\infty \le s - \frac{5}{4}$, $\|w_2\|_\infty \le s - \frac{5}{4}$, $\|w_2 - g\|_2 \le \frac{5}{2}$
C	box of p_0, w_1, and box of w_1, w_2 widened on the inner side only; $y_t = w_2$	$\|w_1\|_\infty \le s$, $\|w_2\|_\infty \le s - 1$, $\|w_2 - g\|_2 \le 3$
algorithm	tube of radius 1 around both legs; $\|y_t - w_2\|_2 \le \frac{1}{2}$	$\|h\|_\infty \le s - 2$, $\|w_1\|_\infty \le s - 1$, $\|w_2\|_\infty \le s - 1$, $\|w_2 - g\|_2 \le \frac{5}{2}$

Every row also carries $6 \le \|w_1 - h\|_\infty \le 99$ in G_1. The agents' rows A and C are less conservative than the algorithm's row because they read the handover code instead of assuming a uniform tube.